

Electric Circuits II Project Report

Prepared by

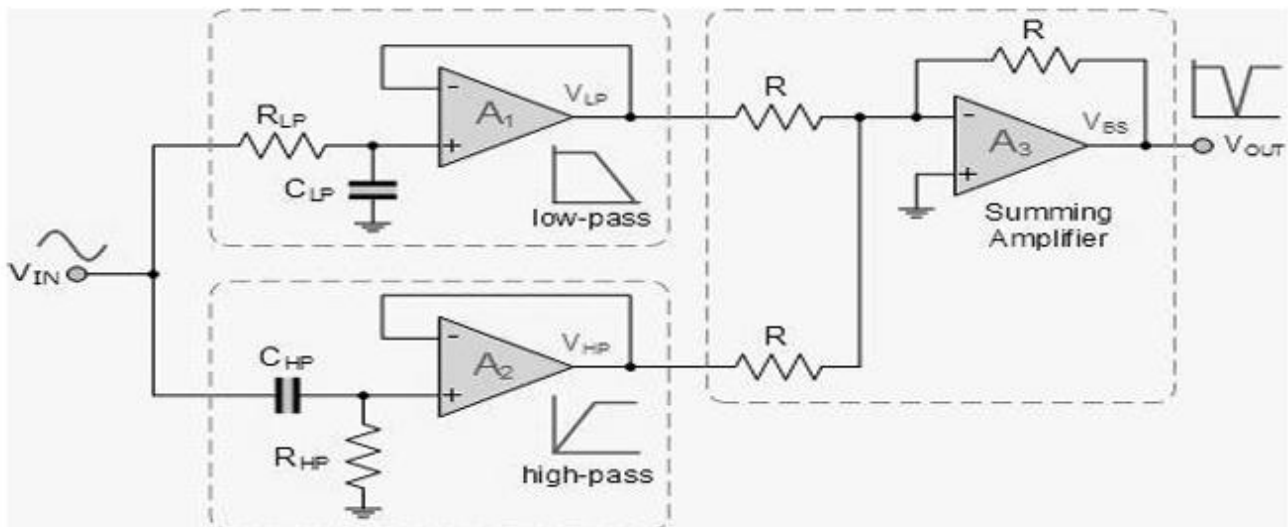
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Tutorial Number: 14

Theoretical background:

Band Stop filter circuit 4 (Basic wideband Band stop filter):

A band-stop filter attenuates signals within a specific frequency range called the stopband while allowing frequencies outside this range to pass with minimal attenuation. For frequencies outside the stopband, the gain is ideally 1 (0 dB).



The circuit consists of three main sections:

1. Low-Pass Filter:

- Components: R_{LP} , C_{LP}
- Defines the upper limit of the low-pass section, allowing signals below the frequency to pass

$$f_L = \frac{1}{2\pi R_{LP} C_{LP}}$$

- RLp: Resistor for low pass.
- CLp: Capacitor for low pass.

2. High-Pass Filter:

- Components: RHp, CHp
- Defines the lower limit of the low-pass section, allowing signals above the frequency to pass.

$$f_H = \frac{1}{2\pi R_{HP} C_{HP}}$$

- RHp: Resistor for high pass.
- CHp: Capacitor for low pass.

3. Summing amplifier:

- Combines the outputs of both filters
- Attenuates frequencies between fL and fH

Key Equations

A. Stopband Width:

- The range between f_L and f_H defines the rejected frequency band.

B. Center Frequency (f_0):

$$f_0 = \sqrt{f_L \cdot f_H}$$

C. Cut off Frequency:

- A specific frequency point where a circuit starts to attenuate the intensity of the signal

D. Gain:

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)}$$

Filter design:

Component Section:

1. Low-Pass Section (sets $f_L = 1.59\text{kHz}$):

- $R_{LP} = 10\text{k}\Omega$
- $C_{LP} = 10\text{nF}$

2. High-Pass Section (sets $f_H = 15.9\text{kHz}$):

- $R_{HP} = 10\text{k}\Omega$
- $C_{HP} = 1\text{nF}$

3. Summing Amplifier:

- Gain resistors (R) = $10\text{k}\Omega$ each.
- Feedback resistor (R) = $10\text{k}\Omega$ for unity gain.

4. Center Frequency f_0 :

- $f_0 = 5.028\text{kHz}$
- $f_L = 1.59\text{kHz}$
- $f_H = 15.9\text{kHz}$

$$f_0 = \sqrt{f_L \cdot f_H}$$

Simulation results:

Provide the screenshots for the obtained simulation results and comments on the simulation results.

You need to run two simulation profiles:

i. AC sweep simulation:

to sketch the gain versus the frequency, where V_{ac} is used as the input signal (One sketch is required).

NB: when you acquire the results from the AC Sweep, you should make sure the gain is ≥ 1

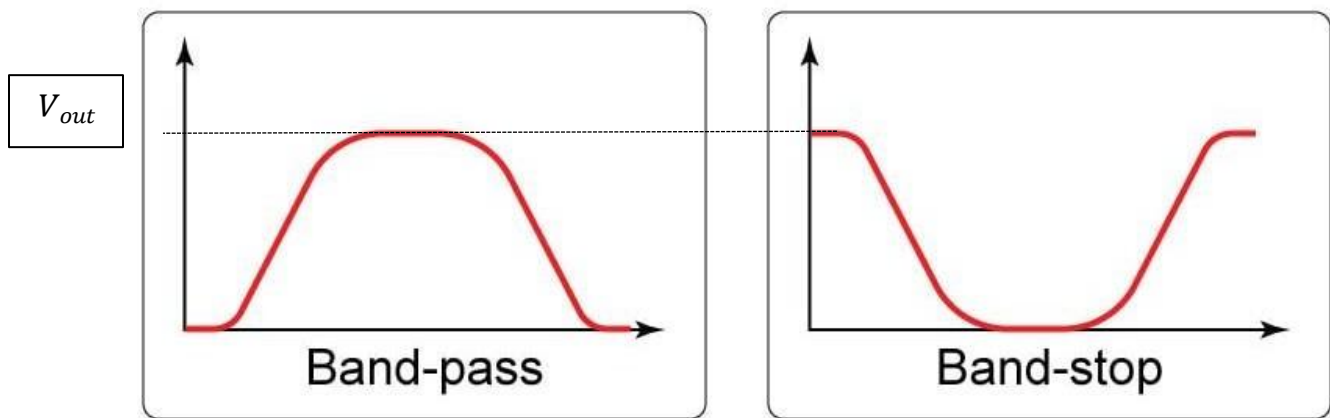


Figure 1 Expected outcome for the AC Sweep

ii. Time domain simulation:

to sketch voltage amplitude versus time, where V_{sin} is used as the input signal at three different frequencies ($f < f_0$, $f = f_0$ and $f > f_0$) (three sketches are required).

Circuit simulation:

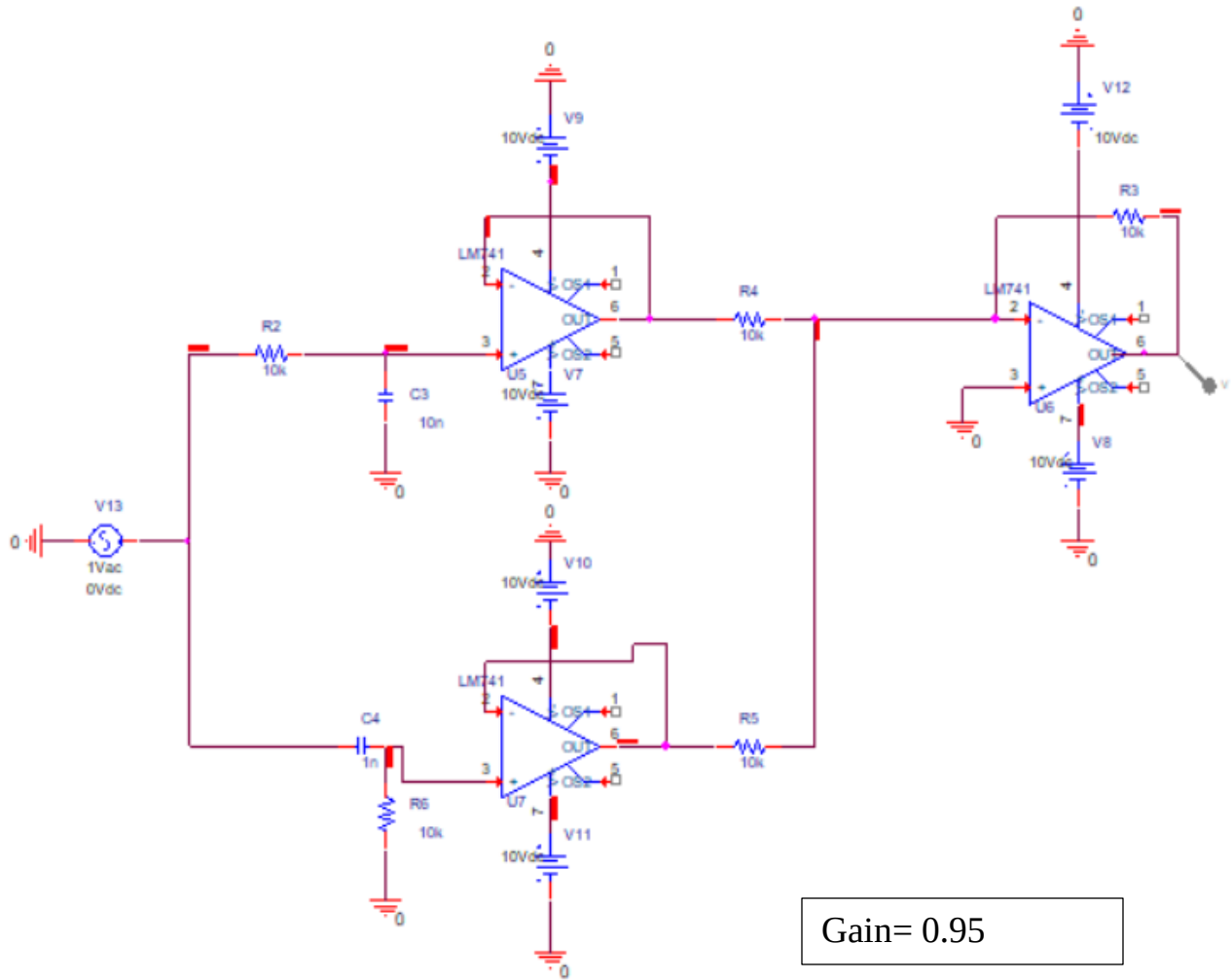
The circuit was constructed using PSpice with the following components:

- 3 Op-Amp: LM741 → from Eval library.
- 5 Resistors: 10 k Ω → from library Analog library.
- Vin (f= 1k \ 8k \ 100k, amp = 2V, Voff \ Vac= 0V) (Time Domain) → from Source library.
- Vac (1V) (Ac Sweep) → from Source library.
- Capacitors: 10 nF (low-pass), 1 nF (high-pass) → from Analog library.
- 6 Vdc (-10V & +10V) → from Source library.
- 10 Ground (0) → from Source library.

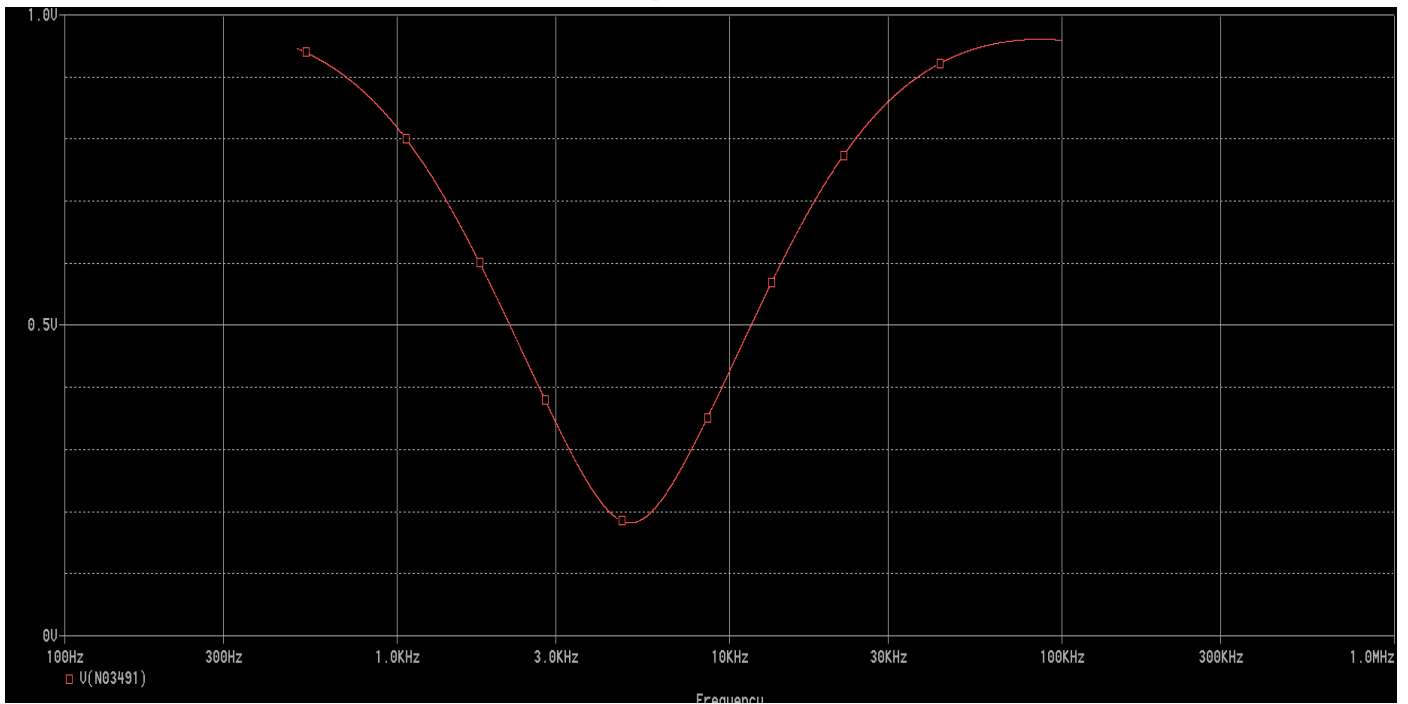
AC Sweep Profile:

General	Analysis Type: AC Sweep/Noise	
Analysis	Options:	
Configuration Files	☒ General Settings ☐ Monte Carlo/Worst Case ☐ Parametric Sweep ☐ Temperature (Sweep) ☐ Save Bias Point ☐ Load Bias Point	
Options	AC Sweep Type: ☐ Linear ☒ Logarithmic Decade	
Data Collection	Start Frequency:	500
Probe Window	End Frequency:	100k
	Points/Decade:	10000
	Noise Analysis ☐ Enabled Output Voltage: I/V Source: Interval:	
	Output File Options ☐ Include detailed bias point information for nonlinear controlled sources and semiconductors (.OP)	
OK Cancel Apply Reset Help		

AC Sweep



Gain= 0.95

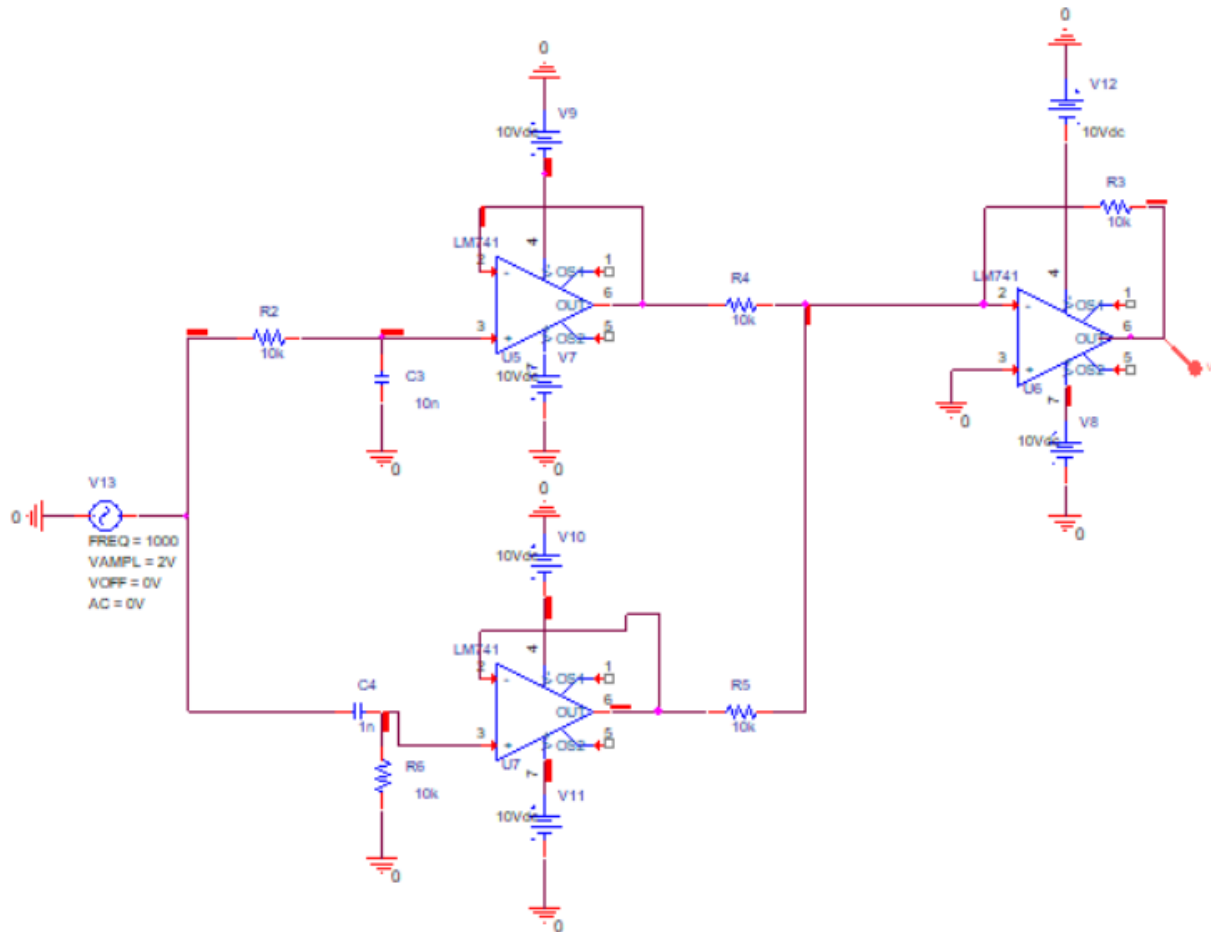


Time Domain Profile:

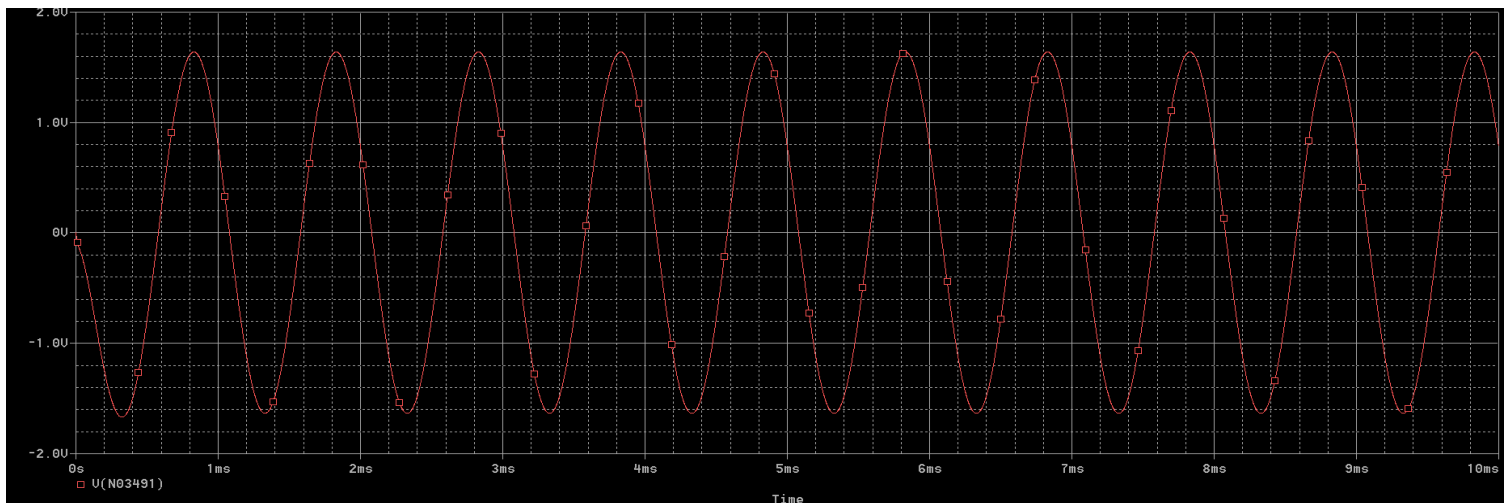
General Analysis Configuration Files Options Data Collection Probe Window	Analysis Type: Time Domain (Transient) ▼	
	Options: ☒ General Settings ☐ Monte Carlo/Worst Case ☐ Parametric Sweep ☐ Temperature (Sweep) ☐ Save Bias Point ☐ Load Bias Point ☐ Save Check Point ☐ Restart Simulation	
Run To Time : 100ms seconds (TSTOP)		Start saving data after : 0 seconds
Transient options: Maximum Step Size 1u seconds ☐ Skip initial transient bias point calculation (SKIPBP)		☐ Run in resume mode Output File Options...

OK Cancel Apply Reset Help

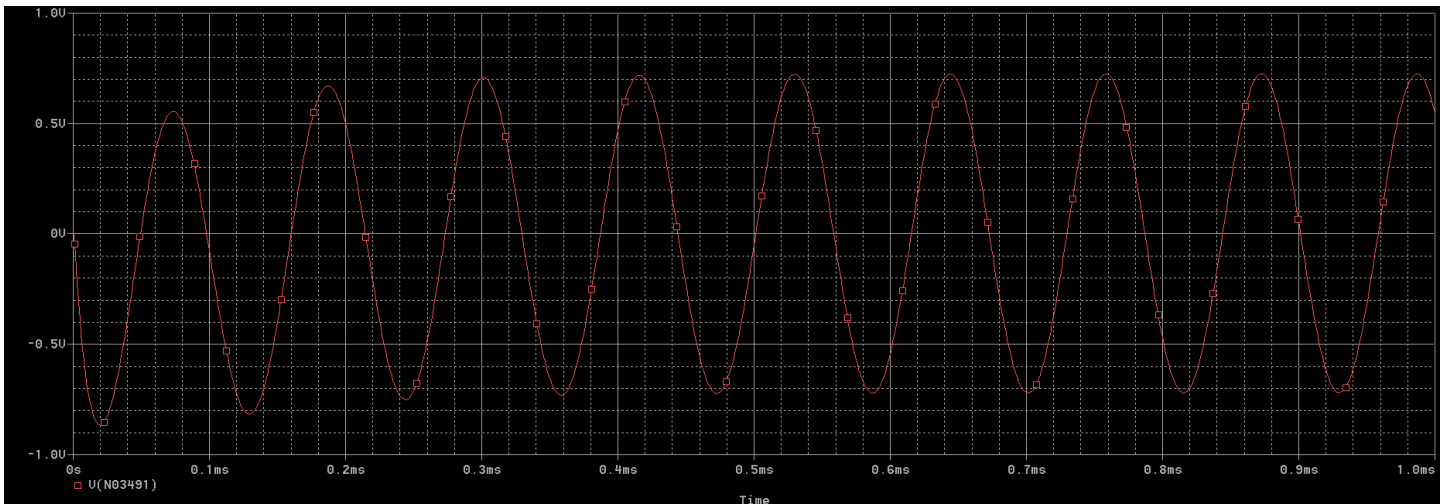
Time Domain



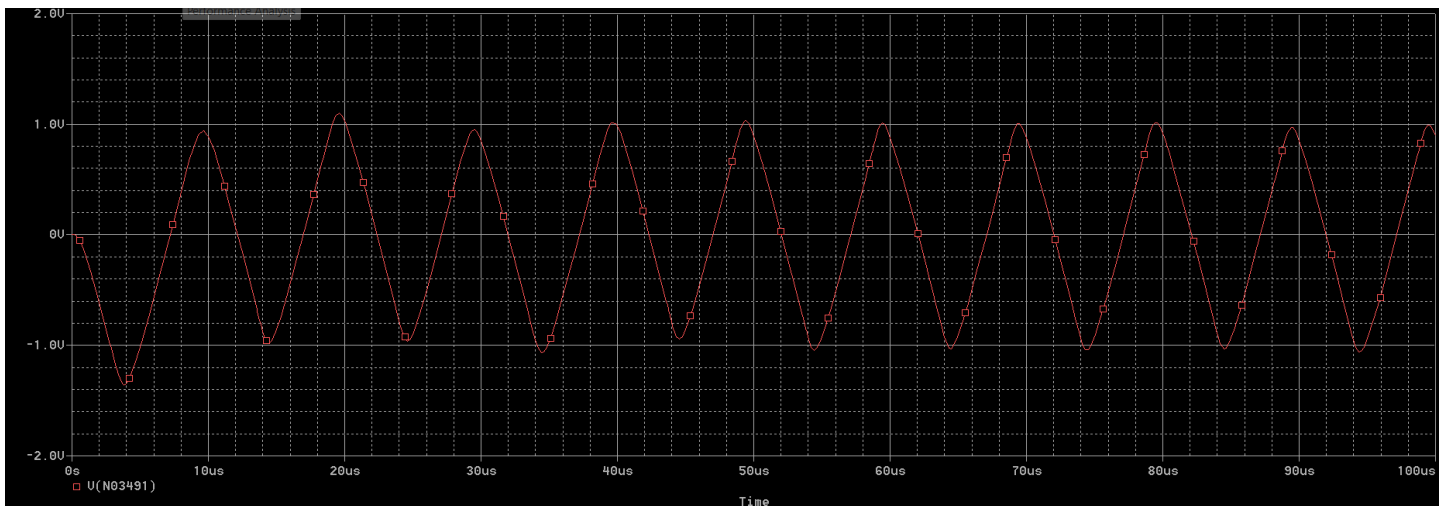
$f_L > f \rightarrow (f=1\text{kHz}) \rightarrow$ (the output is accurate because it is outside the range and circuit is band stop) $V= 1.6\text{V}$



$f_H > f$ and $f_L < f \rightarrow (f = 8.75\text{kHz}) \rightarrow$ (the output is accurate because it is inside the range and circuit is band stop, therefore voltage decreased) $V = 0.68\text{V}$



$f_H < f \rightarrow (f = 100\text{kHz}) \rightarrow$ (the output is accurate because it is outside the range and circuit is band stop) $V = 1\text{V}$



Hardware implementation:

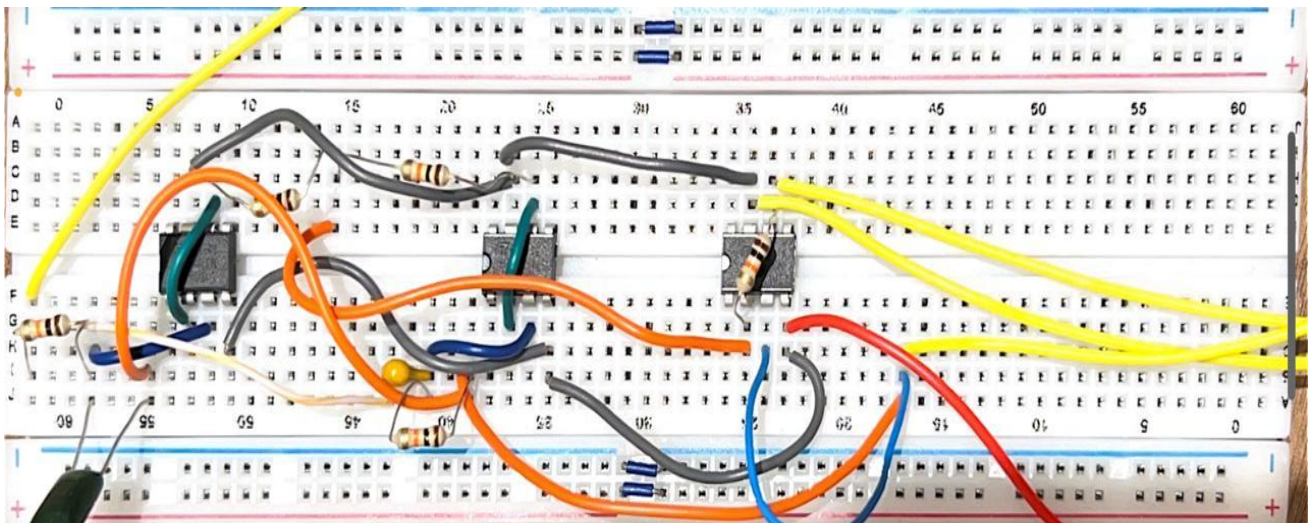
This should include the details of the purchased components and pictures for the hardware implementation of the filter circuit.

Hardware implementation of the filter circuit:

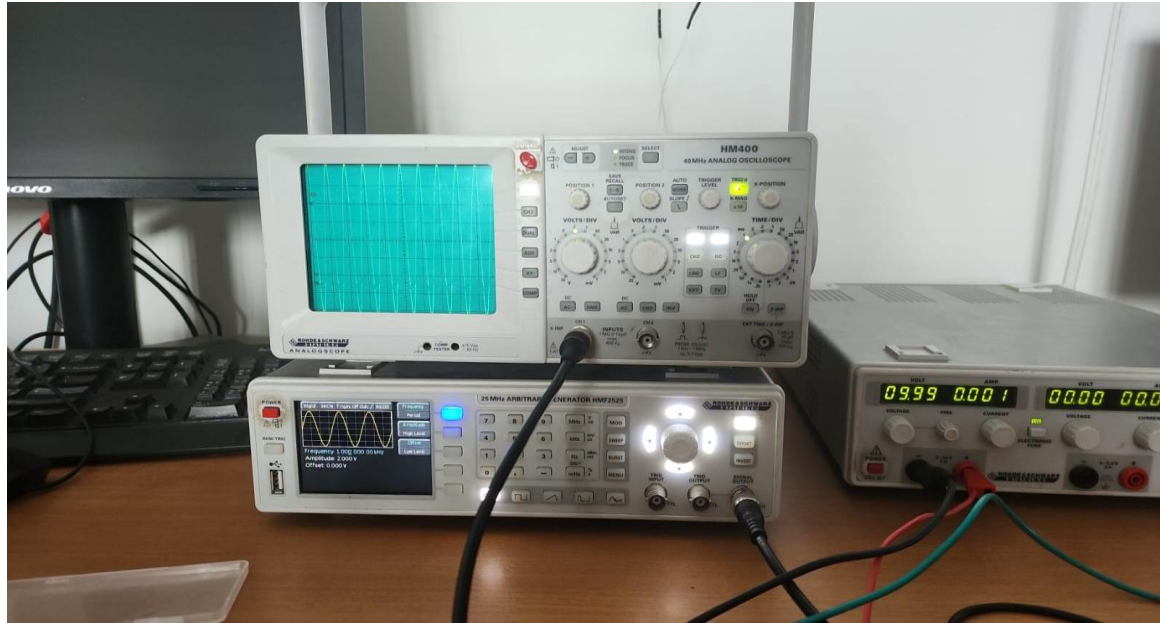
Components Used:

- 3 Op-Amp: LM741
- 5 Resistors: $10\text{k}\Omega$
- Capacitors: 10 nF (low-pass), 1 nF (high-pass)
- Breadboard and connecting wires

Circuit implementation:

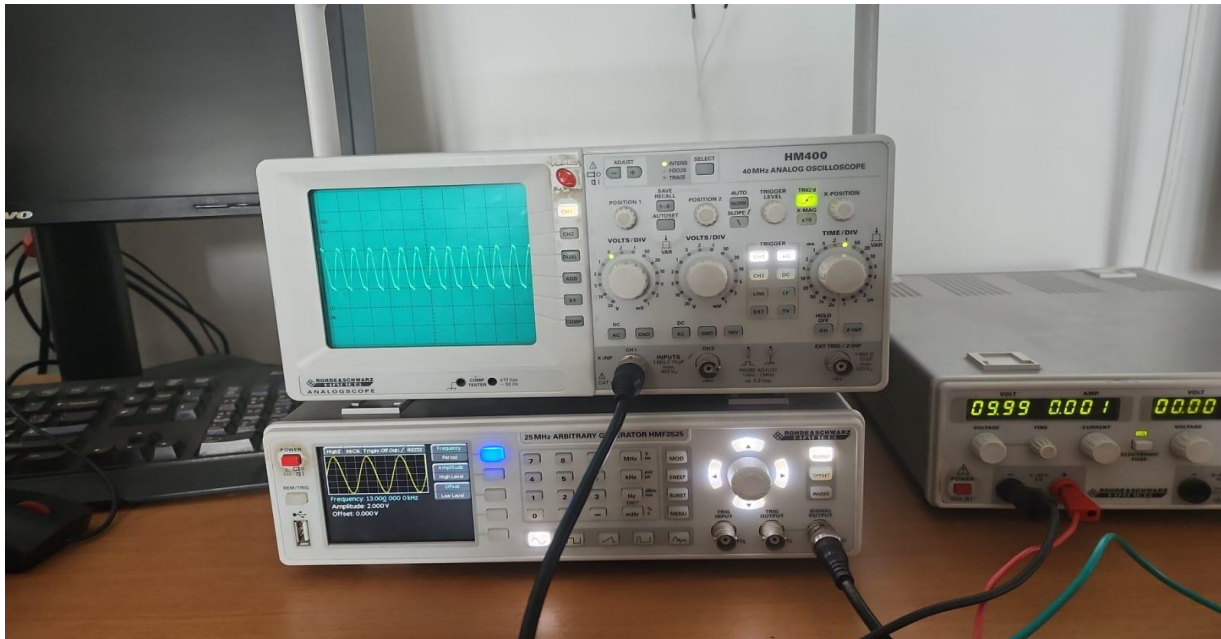


Hardware results:



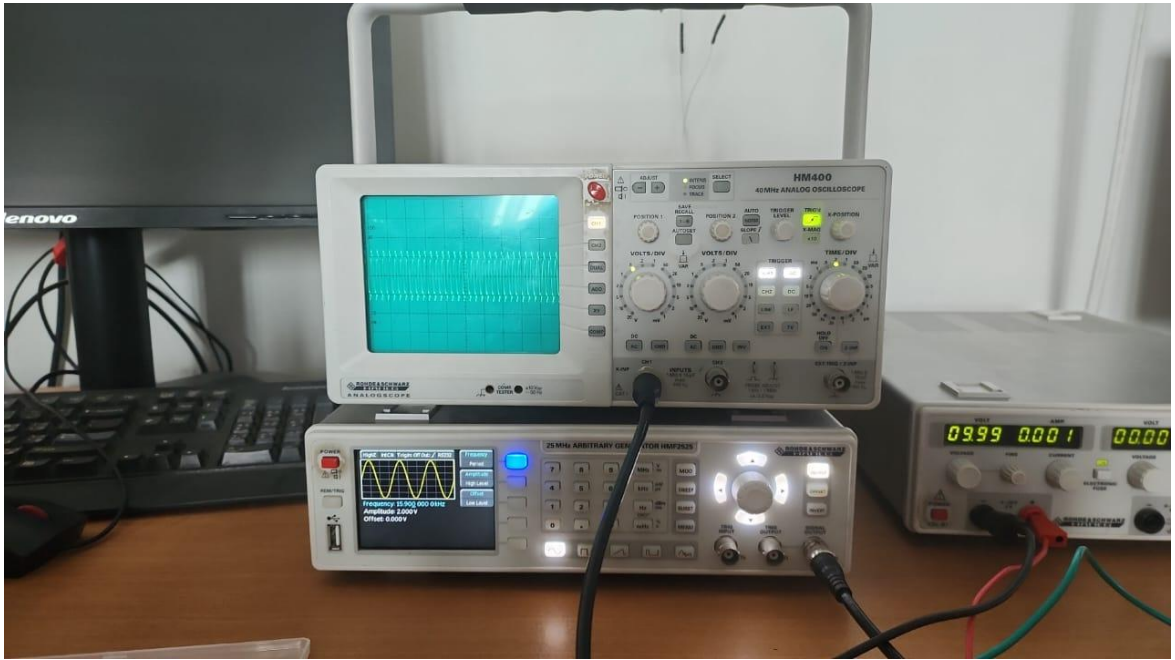
$f < f_0$:

For input signals with frequencies lower than f_0 , the filter operates in its passband, allowing the signal to pass through with minimal alteration. The oscilloscope displays input and output waveforms that are virtually indistinguishable in terms of amplitude and phase.



$f = f_0$:

At the center frequency, the filter operates in its stopband region. This results in severe attenuation of the output signal, which appears drastically reduced or nearly flat when viewed on the oscilloscope relative to the input.



$f > f_0$:

For frequencies above the center frequency (f_0), the filter's passband behavior resumes, with the oscilloscope displaying output signal amplitude like the input.